



Tshwane University of Technology

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COMPUTER SYSTEMS ENGINEERING

MODULE: LOGIC DESIGN 226

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FINAL PROJECT: BLUETOOTH ACCESS CONTROL SYSTEM

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1. Objective

- The objective is to design and implement a secure access control system that unlocks a door or a relay–controlled mechanism when an authorised Bluetooth device is connected, like a smartphone.
- The system should validate the device using its MAC address, provide visual and audible feedback (LEDs and buzzer), and display the access status on an LCD screen.

This ensures:

- Convenience: Hands-free unlocking for authorized users.
- Security: Only preregistered devices can trigger access.
- Feedback: Clear indication of access status through LEDs, buzzer patterns, and display messages.

2. Material Used

- ESP32 Development Board
(Built-in Bluetooth and Wi-Fi capabilities)
- Relay Module
(To control the lock mechanism)
- Buzzer
(For audible feedback)
- LEDs
 1. Green LED (Access Granted)
 2. Red LED (Access Denied)
 3. Yellow LED (Waiting for connection)
- Resistors
(220Ω for LEDs)
- LCD Display
(For displaying access status.)

3. Procedure

Step 1: Hardware Setup.

- Use software like “Wokwi” or Proteus to model the circuit first and test its logic.
- Source the components like the ESP32 board.
- Place the ESP32 board on the breadboard.
- Connect the relay module to the ESP.
- Connect the LEDs to GPIO 19, 22, and 23 via 220Ω resistors.
- Buzzer to GPIO 21.
- LCD to GPIO (4,14,15,25,26,27) for the data, set, and reset pins.

Step 2: Software Setup

- Install Arduino IDE.
- Add the ESP32 board support to the IDE (ESP32 Dev Module).
- Install required libraries like “BluetoothSerial.h” for Bluetooth and “LiquidCrystal.h” for LCD operation.
- Write the code.



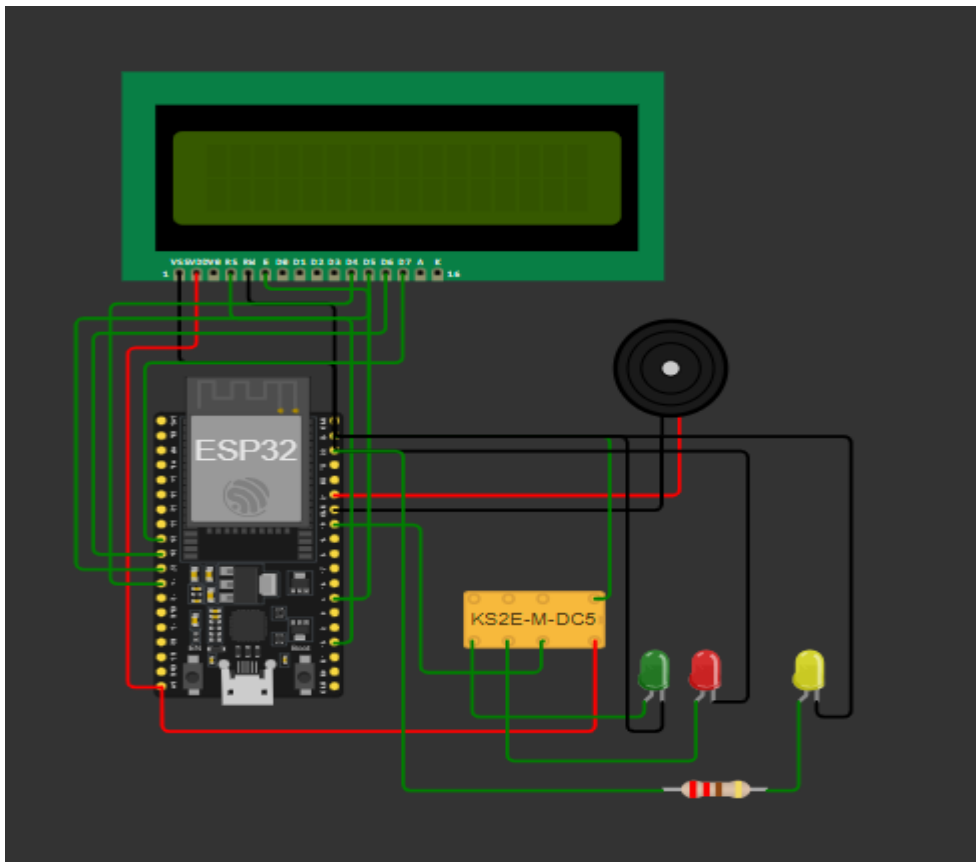
```
1 #include <liquidcrystal.h>
2 #include "BluetoothSerial.h" // Uncomment for actual Bluetooth
3 BluetoothSerial SerialBT;
4
5 // LCD connections: RS, E, D4, D5, D6, D7
6 LiquidCrystal lcd(15, 4, 14, 27, 26, 25);
7
8 // simulated authorized MAC (for testing via Serial Monitor)
9 String authorizedMAC = "AA:BB:CC:DD:EE:FF";
10
11 // Pin definitions
12 const int buzzer = 21;
13 const int waitingLED = 22;
14 const int greenLED = 19;
15 const int redLED = 23;
16
17 void setup() {
18   Serial.begin(115200);
19   SerialBT.begin("ESP32_Access"); // Enable for actual Bluetooth pairing
20
21   // Initialize LCD
22   lcd.begin(16, 2);
23   lcd.print("Access Control");
24   delay(2000);
25   lcd.clear();
26   lcd.print("System Ready");
27
28   // Initialize pins
29   pinMode(buzzer, OUTPUT);
30   pinMode(waitingLED, OUTPUT);
31   pinMode(greenLED, OUTPUT);
32   pinMode(redLED, OUTPUT);
33 }
```

- Compile and download the code to the board.

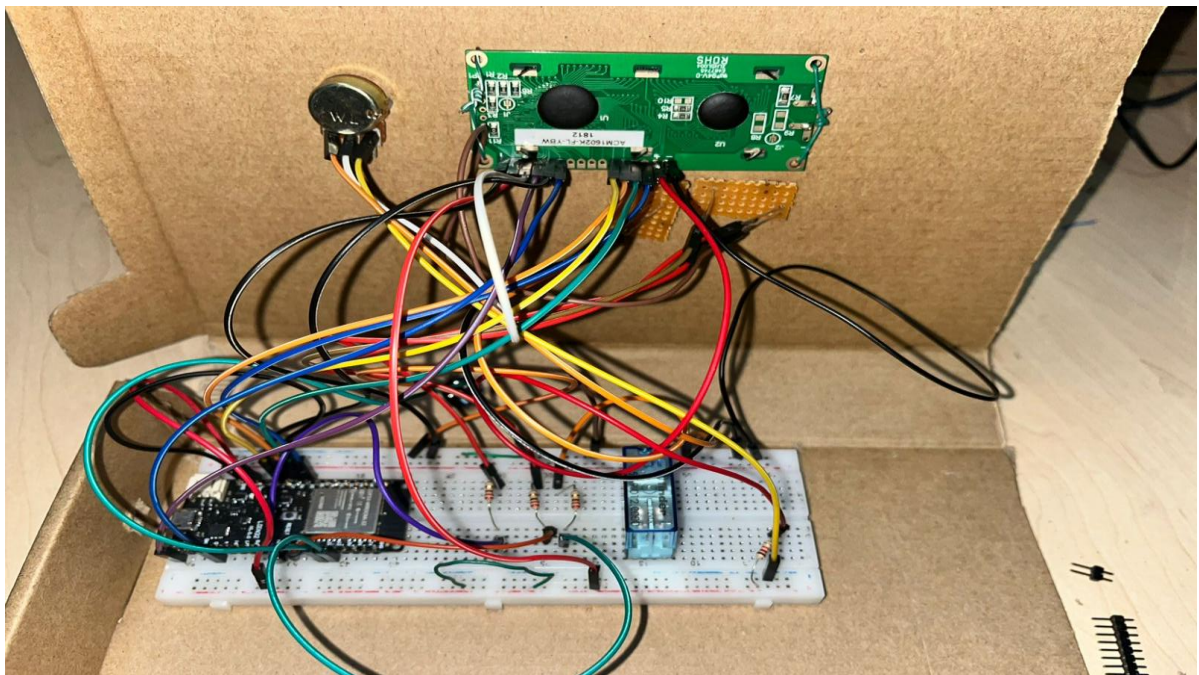
Step 4: Testing.

- Start the board and run the code, while monitoring the serial output.
- Pair an authorised Bluetooth device to test each scenario.
- Verify LCD messages and LEDs for each case.

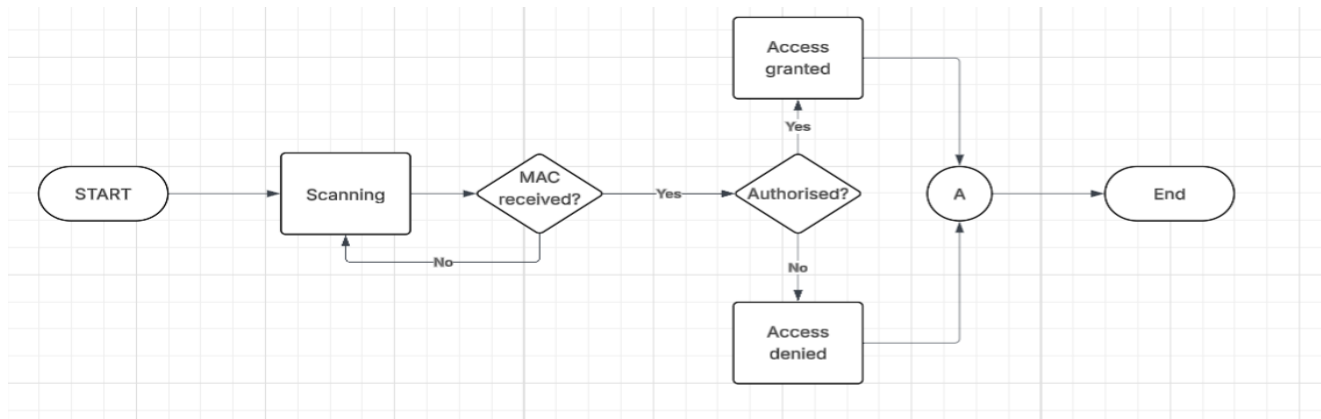
4. Circuit Diagram.



5. Practical Layout



- Flow Chart



5.Results & Observations.

- Truth table

Bluetooth	MAC authorised	Output LED	Buzzer Pattern	LCD Message	Relay
0	X	Waiting	None	System ready	Off
1	0	Red	2 Buzz	Access Granted	Off
1	1	Green	1 Buzz	Access Granted	On

- System Initialization.
 - LCD: Access control then system ready.
 - RED & Green LEDs are off during idle.
- Access granted scenario.
 - Green LED on.
 - Buzzer sound once.
 - LCD: Access Granted.
 - After some time, returns to idle.
- Access Denied Scenario
 - Red LED on.

- Buzzer sounds twice.
 - LCD: Access denied.
 - System returns to idle.
- Performance
 - Bluetooth detection and Verification completed in seconds.
 - LCD updates were clear and responsive.
 - Switching mechanism operated reliably during test.

7. Conclusion.

The Bluetooth-based access control system successfully achieved its primary objective of providing a secure and convenient method of access control using Bluetooth technology. The system demonstrated reliable detection of authorized devices through MAC address verification and provided clear feedback via LEDs, Buzzer, and LCD messages

Key outcomes include:

- Security: Unauthorised devices were consistently denied access, ensuring system integrity.
- Performance: Fast response times and stable operation during multiple test cycles.
- User Experience: Visual and Audio indicators make the whole process simple and intuitive.

This project highlights the potential of IoT and Bluetooth technology for smart security solutions.